```verilog
`timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

module ID_tb();

reg clk;
reg rst_n;
reg [31:0] i_Instr;
reg signed [31:0] i_write_data;
reg [31:0] i_Pc;
reg i_wr;
reg [4:0] i_Rd;
reg flush, i_Boj;

wire [2:0] o_Func3E;
wire [4:0] o_AmoFunct5E;
wire o_IsAmoE;
wire [31:0] o_Rd1E, o_Rd2E, o_ImmE, o_PcE;
wire o_RegSrcE, o_Sel1E, o_Sel2E, o_LoadE, o_BranchE, o_JalE, o_JalrE, o_MemSrcE;
wire [3:0] o_ALUCtrlE;
wire [1:0] o_ResultSrcE;
wire [4:0] o_RdE, a1, a2;
wire [4:0] is_Rs1, is_Rs2;
wire [6:0] is_Opcode;

// Instantiate the ID module
ID uut (
    .clk(clk),
    .rst_n(rst_n),
    .i_Instr(i_Instr),
    .i_write_data(i_write_data),
    .i_Pc(i_Pc),
    .i_wr(i_wr),
    .i_Rd(i_Rd),
    .flush(flush),
    .i_Boj(i_Boj),
    .o_Func3E(o_Func3E),
    .o_AmoFunct5E(o_AmoFunct5E),
    .o_IsAmoE(o_IsAmoE),
    .o_Rd1E(o_Rd1E),
    .o_Rd2E(o_Rd2E),
    .o_ImmE(o_ImmE),
    .o_PcE(o_PcE),
    .o_RegSrcE(o_RegSrcE),
    .o_RdE(o_RdE),
```

```verilog
    .o_Sel1E(o_Sel1E),
    .o_Sel2E(o_Sel2E),
    .o_ALUCtrlE(o_ALUCtrlE),
    .o_LoadE(o_LoadE),
    .o_BranchE(o_BranchE),
    .o_JalE(o_JalE),
    .o_JalrE(o_JalrE),
    .o_MemSrcE(o_MemSrcE),
    .o_ResultSrcE(o_ResultSrcE),
    .is_Opcode(is_Opcode),
    .a1(a1),
    .a2(a2),
    .is_Rs1(is_Rs1),
    .is_Rs2(is_Rs2)
);

// Clock generation
always #5 clk = ~clk;

initial begin
    $display("Starting ID testbench...");
    clk = 0;
    rst_n = 0;
    flush = 0;
    i_Boj = 0;
    i_wr = 0;
    i_Rd = 0;
    i_write_data = 32'd0;

    #10 rst_n = 1;

    // --------- Load x1 with 10 ---------
    i_wr = 1;
    i_Rd = 5'd1;
    i_write_data = 32'd10;
    i_Instr = {12'd0, 5'd1, 3'b000, 5'd0, `I};  // addi x0, x1, 0
    i_Pc = 32'h00000000;
    #10;

    // --------- Load x2 with 20 ---------
    i_Rd = 5'd2;
    i_write_data = 32'd20;
    i_Instr = {12'd0, 5'd2, 3'b000, 5'd0, `I};  // addi x0, x2, 0
    i_Pc = 32'h00000004;
    #10;
```

```
    // --------- AMO ADD (x3 = x1 + x2) ---------
    i_wr = 0;
    i_Instr = {5'b00000, 5'd2, 5'd1, `AMO_W, 5'd3, `AMO};  // amoadd.w x3, x1, x2
    i_Pc = 32'h00000008;
    #10;

    $display("---- Final Results ----");
    $display("o_IsAmoE    = %b", o_IsAmoE);
    $display("o_AmoFunct5 = %b", o_AmoFunct5E);
    $display("o_Rd1E      = %d", o_Rd1E);
    $display("o_Rd2E      = %d", o_Rd2E);
    $display("o_RdE       = %d", o_RdE);
    $display("o_Func3E    = %b", o_Func3E);
    $display("o_ImmE      = %d", o_ImmE);
    $display("a1 (rs1)    = %d", a1);
    $display("a2 (rs2)    = %d", a2);
    $display("------------------------");

    #10;
    $finish;
end

endmodule
```